



NorthStar Retail Platform

Microservices-based Retail Platform

– Saint Deemene

The Overview



NorthStar, a retail-style bookstore platform using microservices architecture. The domain is intentionally kept simple—books, stores, inventory, and orders—so we can focus on architecture, service boundaries, and operational concerns rather than business complexity.

Microservices

- Clear service ownership and independent deployment.
- Each service owns its data (separate PostgreSQL per service)
- Platform team can support multiple teams. Store, Order etc

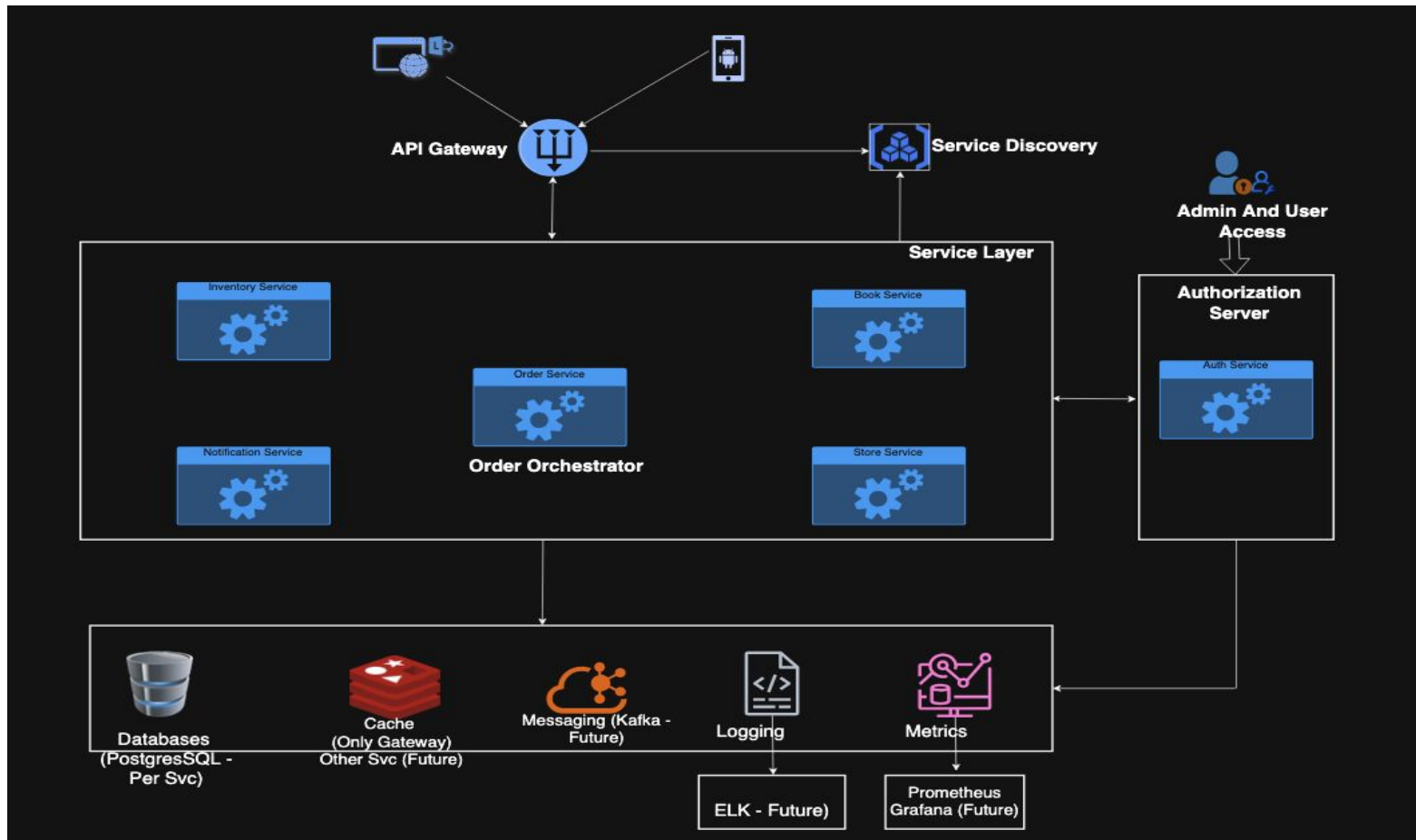
Design Principles

- Service discovery (Eureka) and API Gateway for routing
- The gateway is the single entry point.
- Security with OAuth2/JWT authentication

Domain Simplicity

- Book Service: product catalog
- Store Service: Store master data.
- Inventory Service: store per (store, book)
- Order Service: orchestration and order lifecycle
- Notification Service: for order messaging.

High-Level Architecture



Architecture Decision

DB Per Service

- Clear Ownership
- Independent Deployment

Gateway + Eureka

- Central routing with single entry point
- Apply circuit breaker and retry logic
- Manage service availability

Order Svc Orchestrates Other Services

- The core business flow
- Coordinate validation and reservation



Core Business Flow - Order Creation

Client Request

- **POST-**
``/api/order/v1/orders``
`{ "storeId": "{{store_uid}}",
 "lineItems": [{ "bookId":
 "{{book_uid}}", "quantity": 2 }] }`
- **Request Includes user JWT token**

Gateway Processing

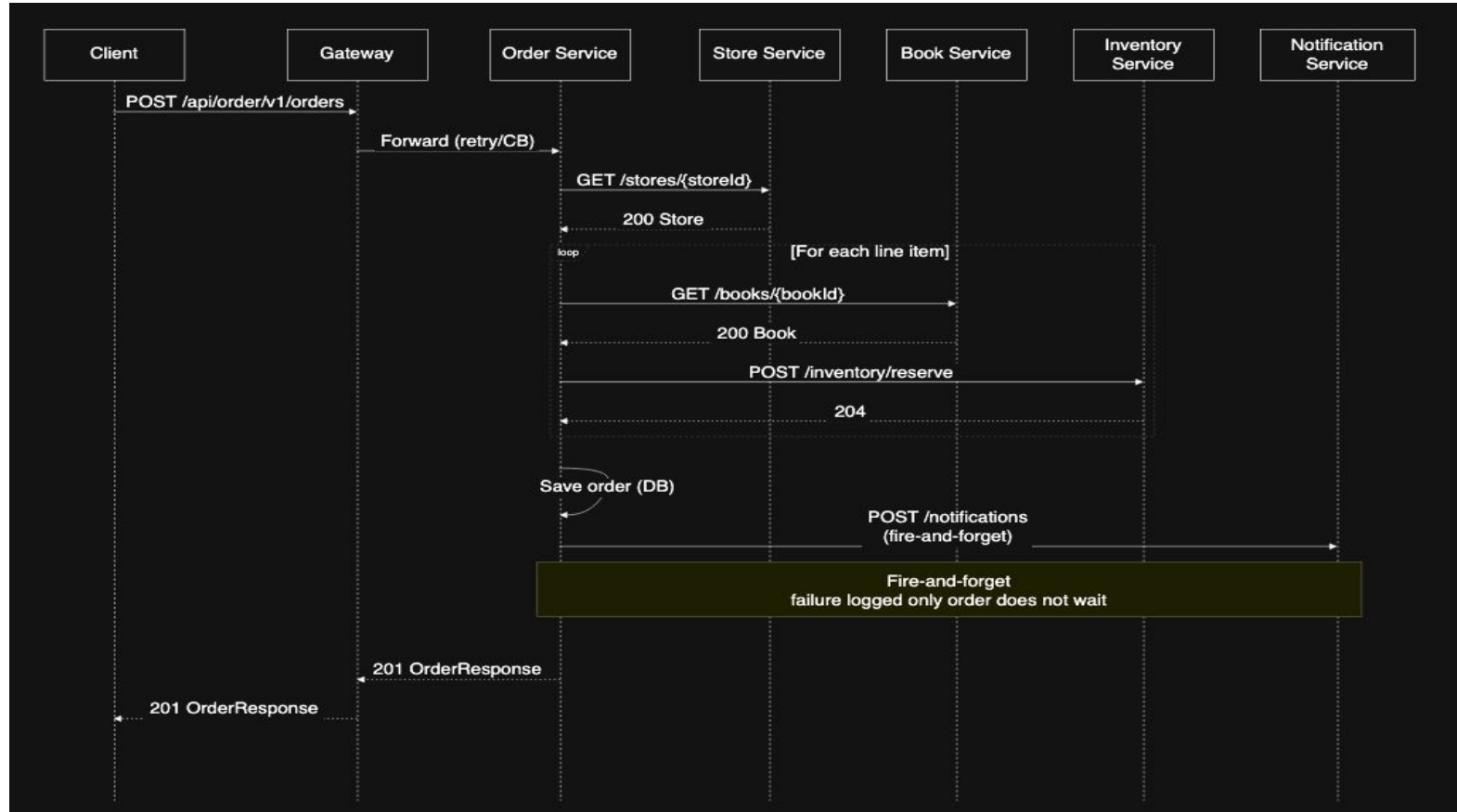
- Routes to Order Service via Eureka (``lb://order-service``)
- Applies circuit breaker and retry logic
- Ensure service availability

Order Orchestration

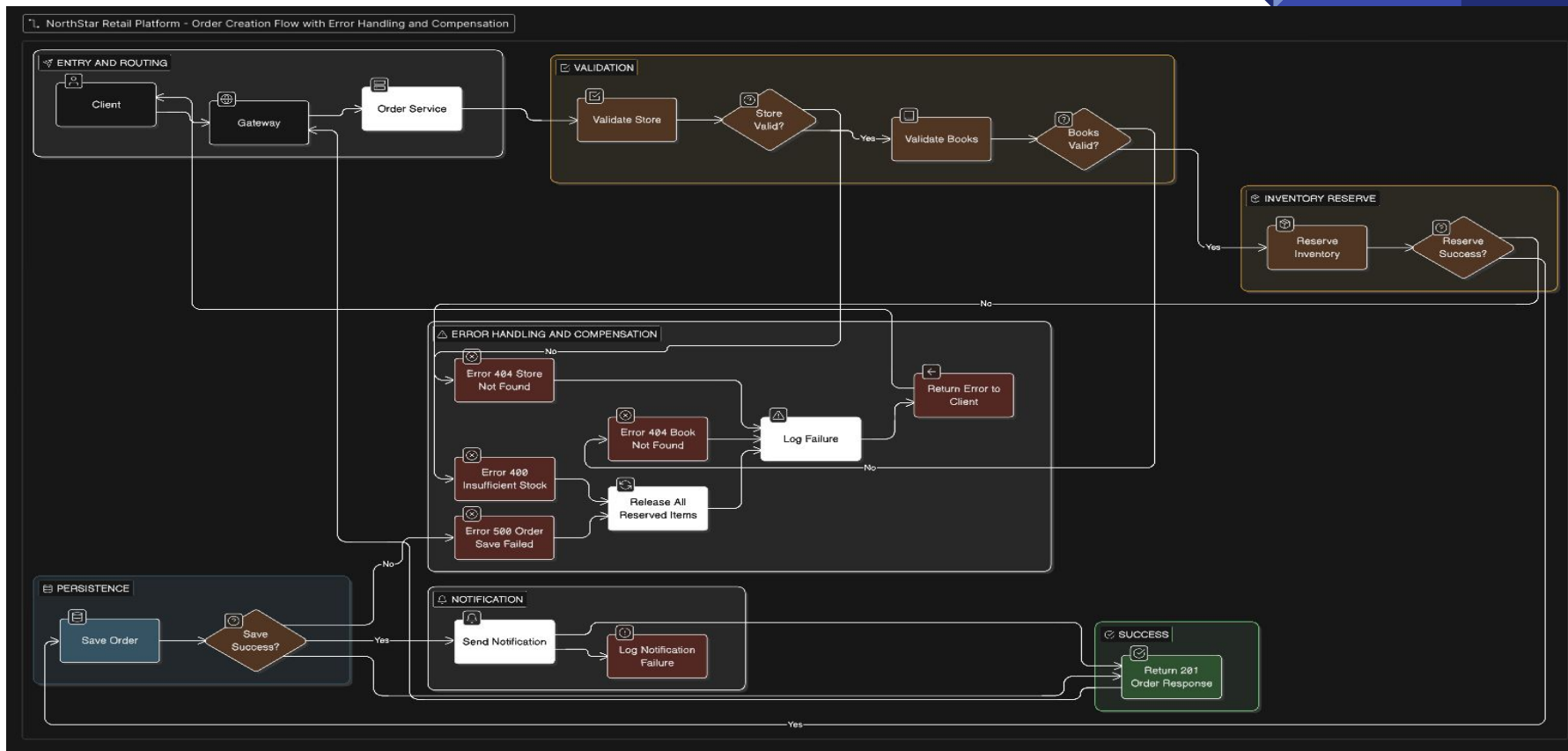
- Extract userId
- Validate Store exists
- For each line item
 - Validate book exists
 - Compute total Order amount
 - Reserve Inventory
- Persist the Order
- Call Notification Service
- Return 201 Order Response to Client



Order Creation Sequence Diagram



Implementation Flow Diagram



NorthStar Retail Platform - Run results

Ran today at 05:50:49 PM

View all runs

Source	Environment	Iterations	Duration	All tests	Errors	Avg. Resp. Time
Runner	NorthStar Local	1	2s 734ms	0	0	228 ms

All Tests

Passed (0)

Failed (0)

Skipped (0)

Errors (0)

Console Log

View Summary

Iteration 1

POST

Flows / Complete Flow / 1. Register Admin

http://localhost:8181/api/auth/v1/guests/admin

201

No tests found

POST

Flows / Complete Flow / 2. Register User

http://localhost:8181/api/auth/v1/guests/user

201

No tests found

POST

Flows / Complete Flow / 3. Login Admin

http://localhost:8181/api/auth/oauth2/token

200

No tests found

POST

Flows / Complete Flow / 4. Login User

http://localhost:8181/api/auth/oauth2/token

200

No tests found

POST

Flows / Complete Flow / 5. Create Book (Admin)

http://localhost:8181/api/book/v1/books

201

No tests found

POST

Flows / Complete Flow / 6. Create Store (Admin)

http://localhost:8181/api/store/v1/stores

201

No tests found

POST

Flows / Complete Flow / 7. Create Inventory (Admin)

http://localhost:8181/api/inventory/v1/inventory

201

No tests found

POST

Flows / Complete Flow / 8. Create Order (User)

http://localhost:8181/api/order/v1/orders

201

No tests found

GET

Flows / Complete Flow / 9. Get Notifications by User

http://localhost:8181/api/notification/v1/notifications/user/cd52ff33-ed2d-4fc3-ab9...

200

No tests found

1 POST NorthStar Retail Plat... / ... / 8. Create Order (User) X

Response

Headers

Request

201

680 ms

881 B

Pretty

1 {

2 "uid":

3 "247bf9b-8798-47af-913e-79d906832e

4 cd",

5 "storeId":

6 "a91d5a96-e0ed-4662-bcb5-3f17039d28

7 47",

8 "status": "PENDING",

9 "amount": 59.98,

10 "createdAt": "2026-02-10T17:50:51.

11 757851",

12 "lineItems": [

13 {

14 "uid":

15 "81e4090e-ab57-404c-937c-97

16 5a77781770",

17 "bookId":

18 "60e00e44-4cce-4a89-b7bb-95

19 4422baded1",

20 "quantity": 2

21 }

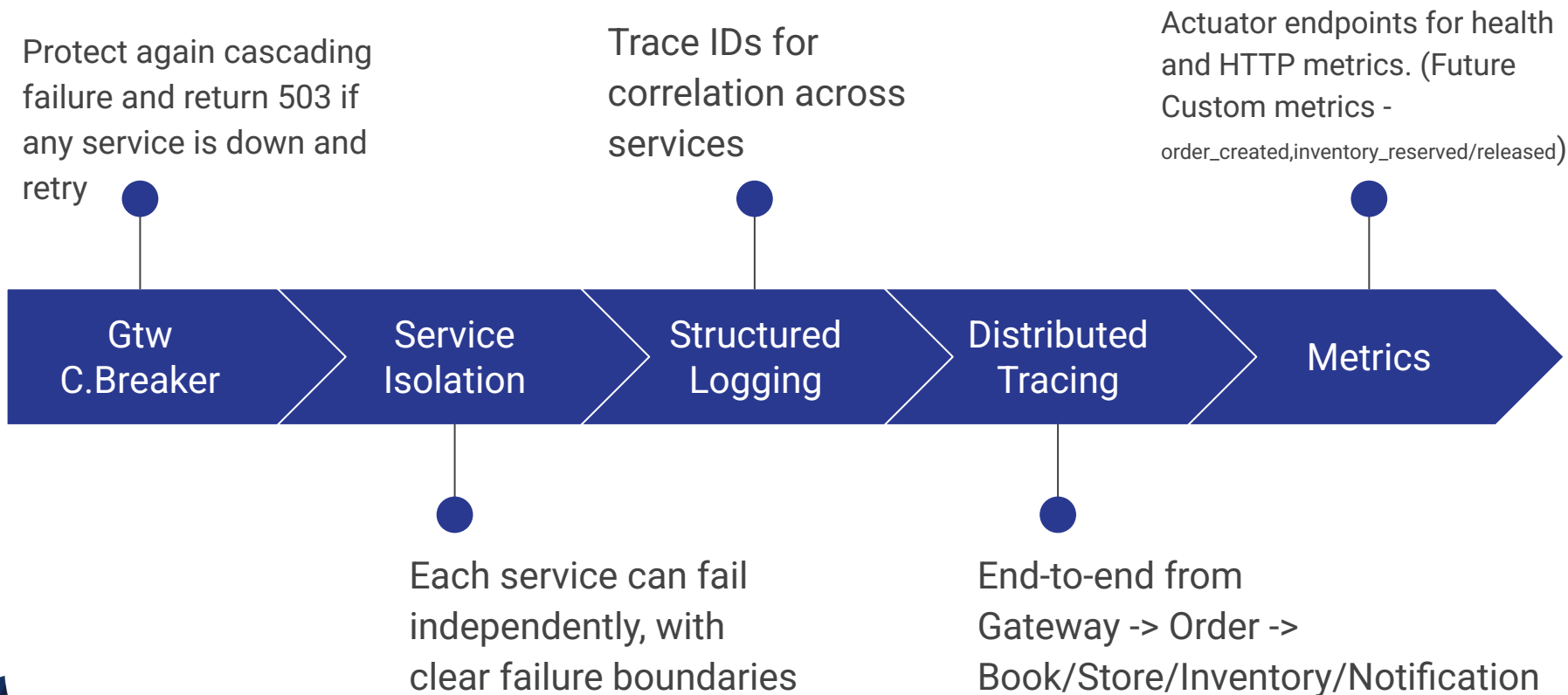
22]

23 }

App Demonstration



Operational Reliability

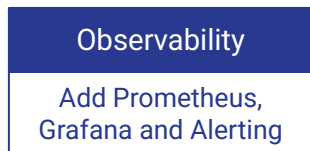


L3 Support

“The system is designed to degrade gracefully. “Logs, metrics, and structured tracing support L3 incident handling e.g, why an order failed and postmortems, . Runbooks are documented per service.



Enhancement



Conclusion

I designed the NorthStar Retail Platform to demonstrate a production-ready microservices architecture with clear service boundaries, operational usage, and graceful degradation. The platform is designed for scale, maintainability, and fast incident response. Each service has clear ownership, making it suitable for a platform team supporting multiple product teams.

